

End to End Encrypted IoT with Dart & Flutter

Flutter Silicon Valley Meetup - October 2022





A brief introduction



Engineer, Atsign & Dart GDE
building a platform that puts people in
control of their data.

Co-host, Tech Debt Burndown Podcast

Cloud Editor, InfoQ

Blogger blog.thestateofme.com

chris.swanz.net for links to socials etc.



@cpswan



Agenda

- Why?
- Flutter Front End
 - ◆ Gauges
- Dart Device Driver
- Connectivity, Security, Privacy
- Review

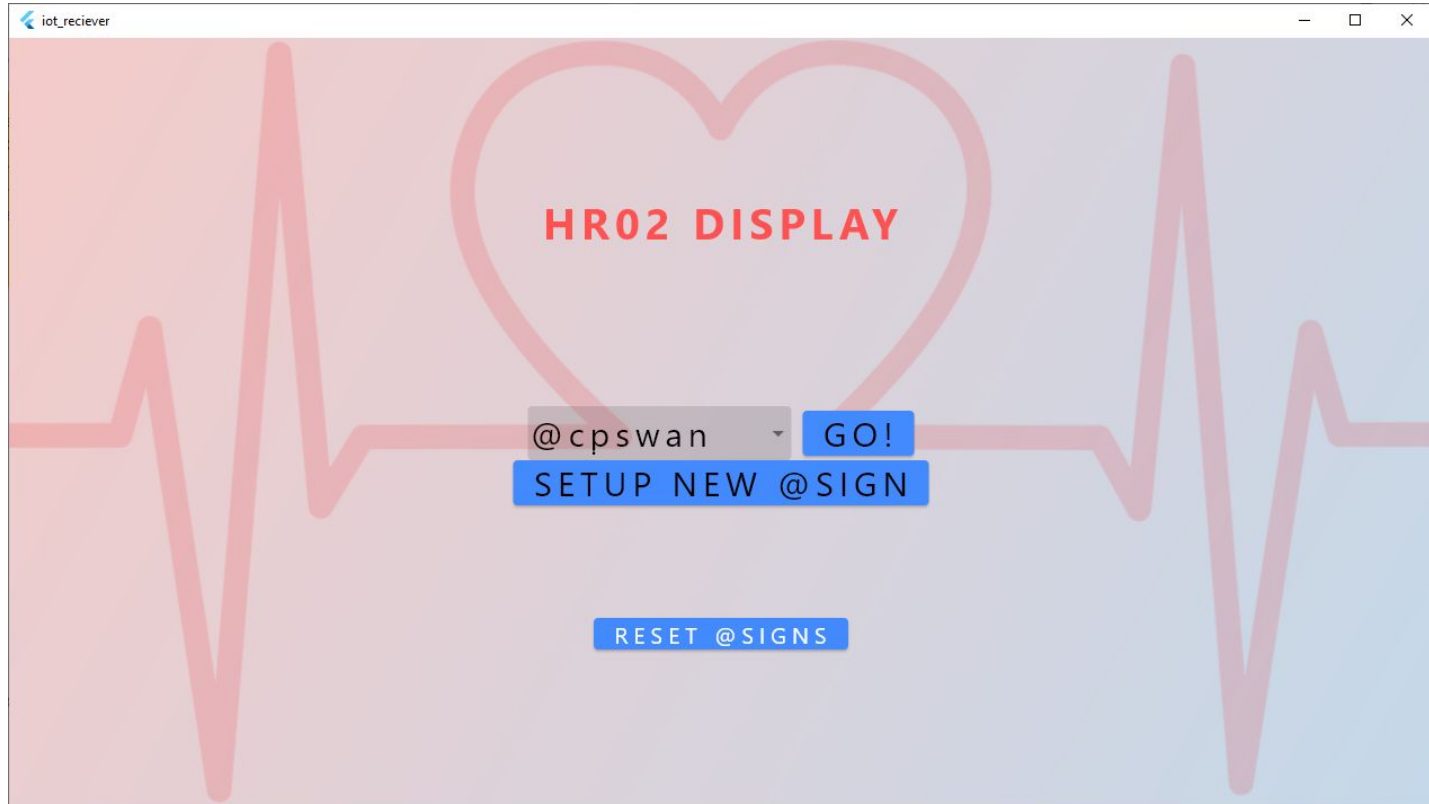


Why?

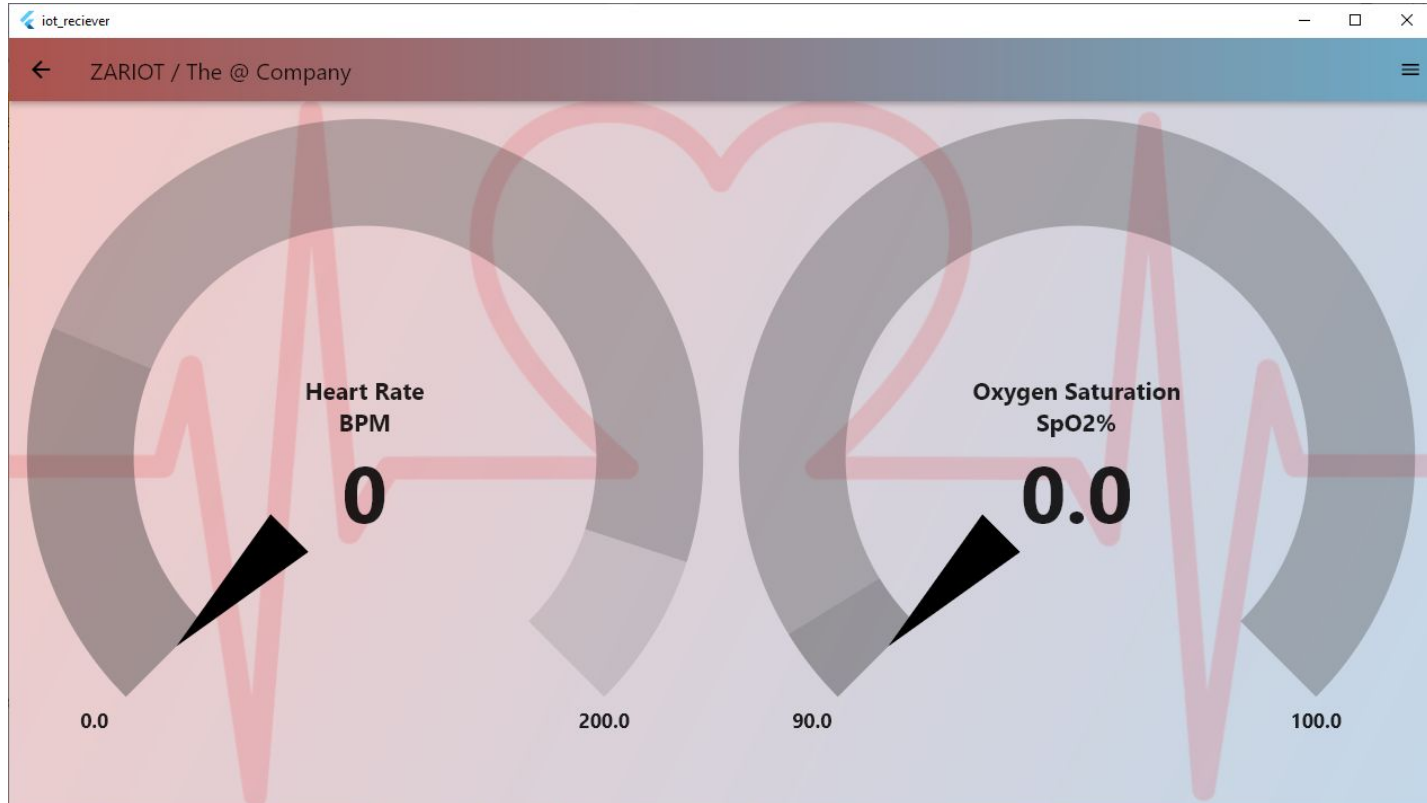
Your Thing - Your Data

Flutter Front End

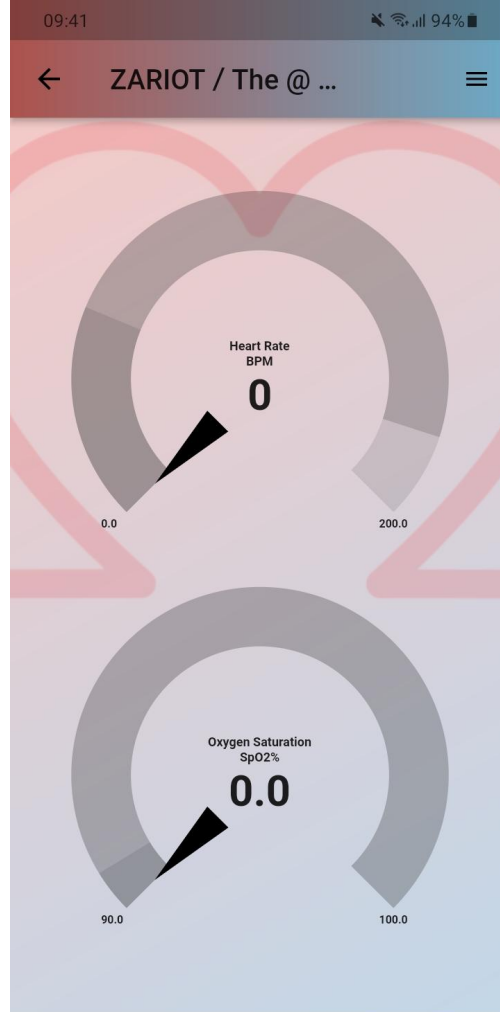
Sign in (Windows Desktop)



Main screen (Windows Desktop)



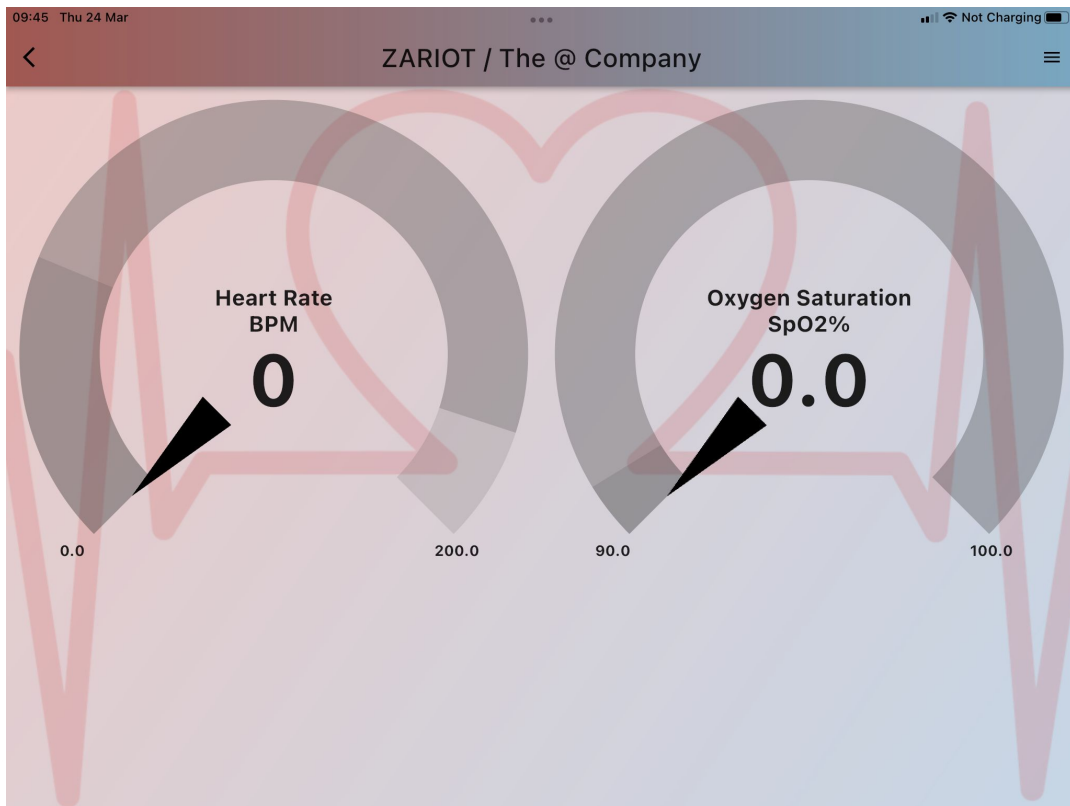
And Android



<https://play.google.com/store/apps/details?id=com.atsign.hro2>



And iOS



<https://apps.apple.com/gb/app/hro2/id1610811005>

New Gauges For Better Visualisation

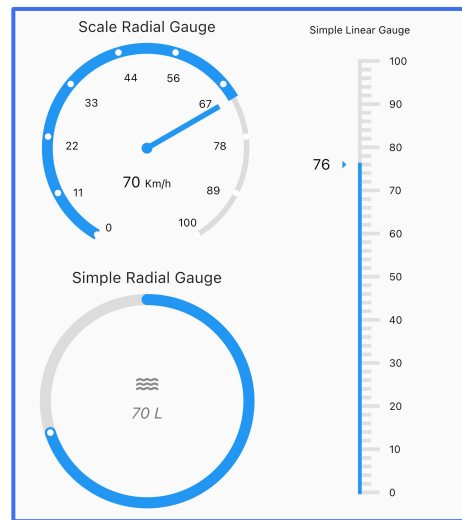
Simple data visualization for IoT Flutter apps

Problem

IoT Flutter apps are being built but there aren't many open source options for dynamic data visualization.

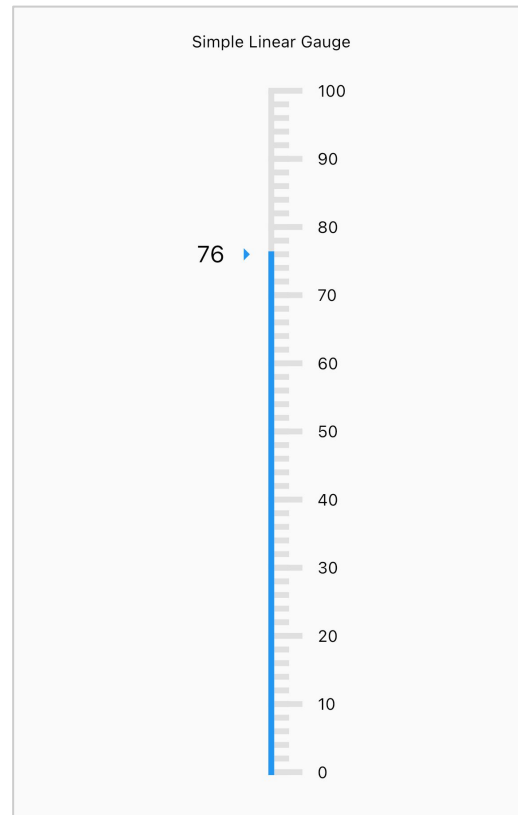
Solution: at_gauges

- Radial gauges
- Vertical and horizontal linear gauges
- Built in Flutter and Dart
- Open Source

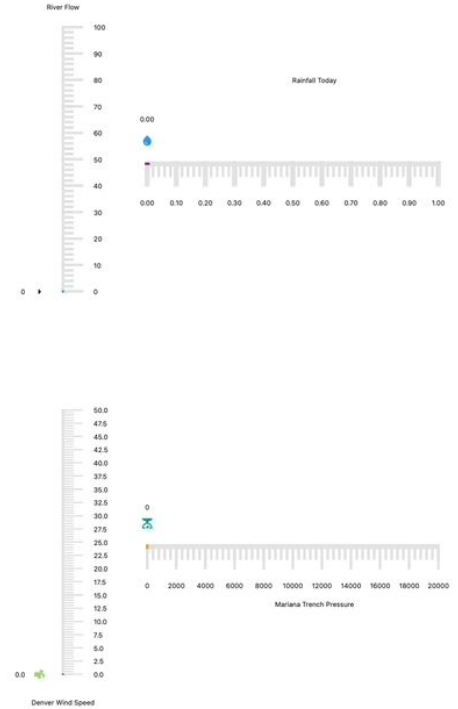
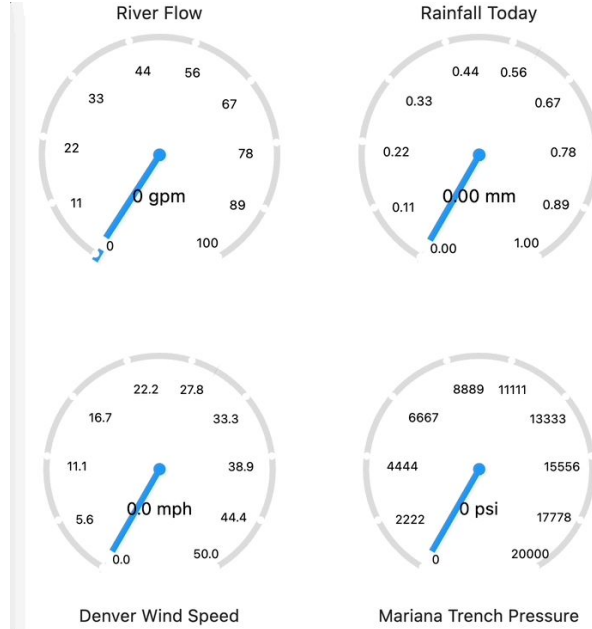
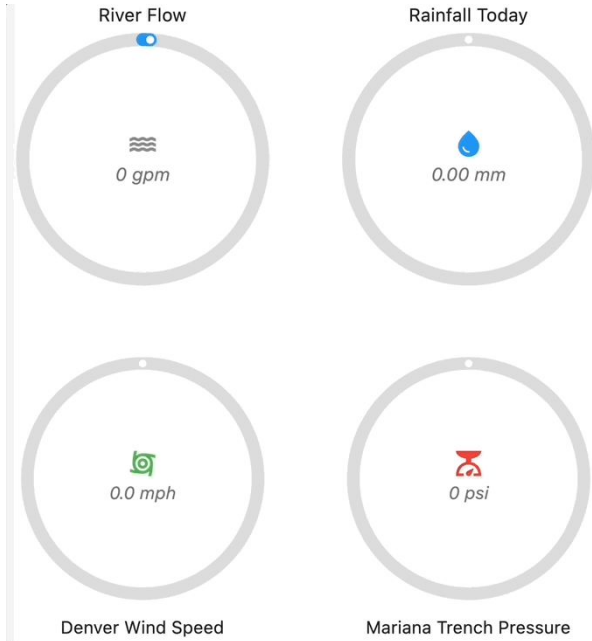


Simple customization

```
SimpleLinearGauge(  
  maxValue: 100,  
  actualValue: 76,  
  //Optional Parameters  
  minValue: 0,  
  divisions: 10,  
  size: double.infinity,  
  title: const Text('Simple Linear Gauge'),  
  titlePosition: TitlePosition.top,  
  pointerColor: Colors.blue,  
  pointerIcon: const Icon(Icons.arrow_right, size: 40),  
  gaugeOrientation: GaugeOrientation.vertical,  
  decimalPlaces: 0,  
  isAnimate: true,  
  animationDuration: 2000,  
  gaugeStrokeWidth: 5,  
  rangeStrokeWidth: 5,  
  majorTickStrokeWidth: 5,  
  minorTickStrokeWidth: 5,  
  actualValueTextStyle: const TextStyle(color: Colors.black, fontSize: 20),  
  majorTickValueTextStyle: const TextStyle(color: Colors.black),  
),
```



at_gauges in action...



Check out the repo!



Github Repo URL

https://github.com/at-sign-foundation/at_gauges

Developed by



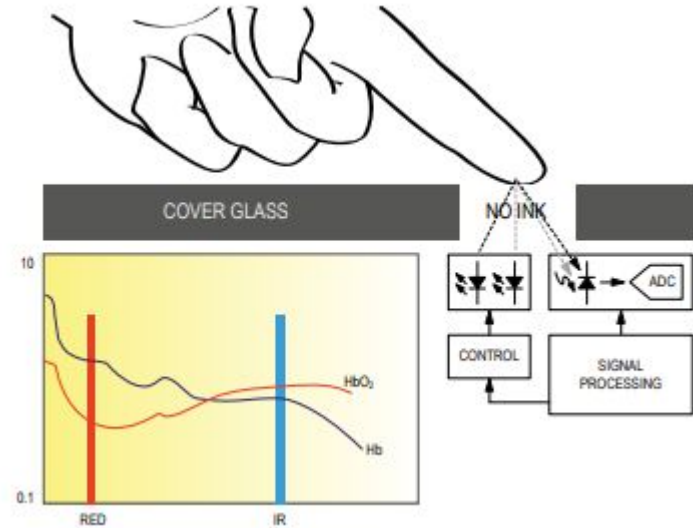
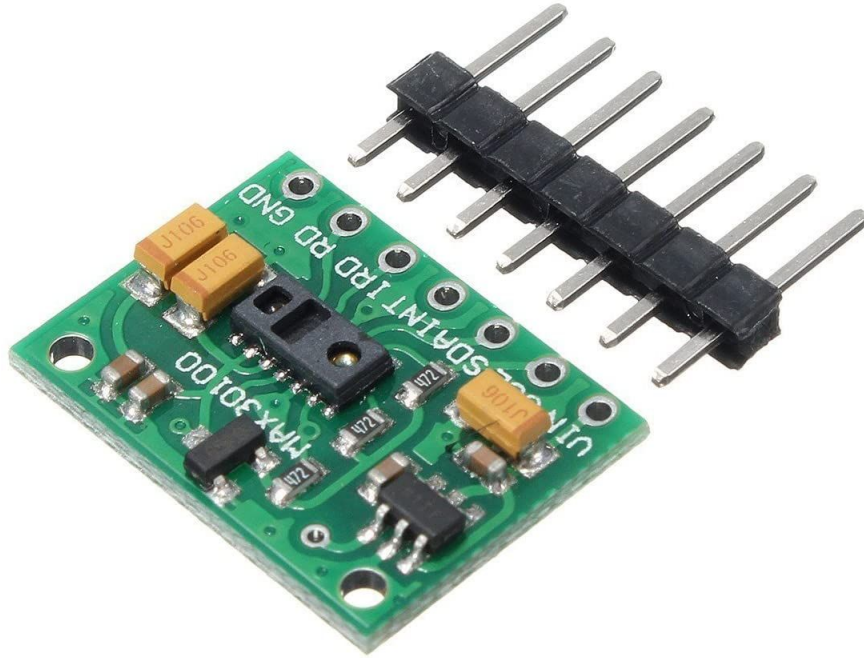
Curtly Critchlow

<https://github.com/curtlycritchlow>



Dart Device Driver

MAX30100 sensor - heart rate and O2 saturation



The programmer's lament

*“We do these things,
not because they are easy,
but because we thought they would be easy”*



Newer sensor - better drivers?



<https://shop.pimoroni.com/products/max30101-breakout-heart-rate-oximeter-smoke-sensor>



C++ (and bits of Python) driver ported to Dart

max30101 1.0.0-RC3 

Published 12 days ago • [atsign.org](#) Null safety

SDK

DART

FLUTTER

PLATFORM

ANDROID

IOS

LINUX

MACOS

WINDOWS

 0

[Readme](#)

[Changelog](#)

[Example](#)

[Installing](#)

[Versions](#)

[Scores](#)

[Admin](#) 

[Activity log](#) 

max30101

A Dart package (developed for use on a Raspberry Pi 4) which interfaces via I2C with the [Max30101](#) sensor and produces illustrative heart rate and O2 saturation readings.

- Primarily based on Raivis Strogonovs' excellent [C++ library](#) which implements heart rate and O2 saturation readings for the Max30100 sensor. (See also his superb accompanying [tutorial](#)). This package is a pretty faithful port to Dart of that C++ code, and also includes the various adjustments (registers, settings, etc.) required to make it work with the Max30101 sensor.
- The package's approach to register interaction was inspired by Philip Howard's [Python library](#) which implements heart rate readings for the Max30105 sensor
- The essential ingredient of I2C support is provided by the https://pub.dev/packages/dart_periphery package



<https://pub.dev/packages/max30101>



We stand on the shoulders of giants

dart_periphery 0.8.29

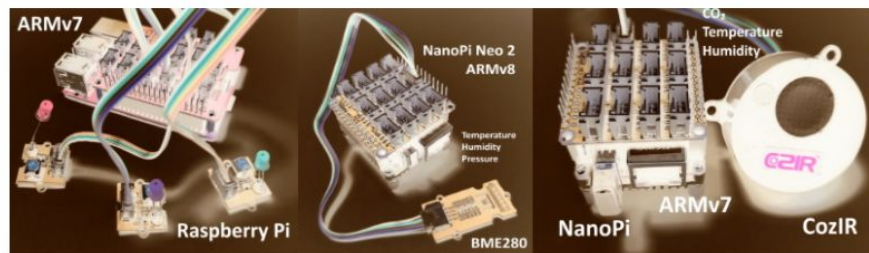
Published 4 days ago • flutterdev.at (Null safety)

SDK | DART | FLUTTER | PLATFORM | ANDROID | IOS | LINUX | MACOS | WINDOWS

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[Readme](#) | [Changelog](#) | [Example](#) | [Installing](#) | [Versions](#) | [Scores](#)

dart_periphery



pub v0.8.29

Introduction

dart_periphery is a Dart port of the native [c-periphery library](#) for Linux Peripheral I/O (GPIO, LED, PWM, SPI, I2C, MMIO and Serial peripheral I/O). This package is specially intended for SoCs like Raspberry Pi, NanoPi, Banana Pi et al.



https://pub.dev/packages/dart_periphery

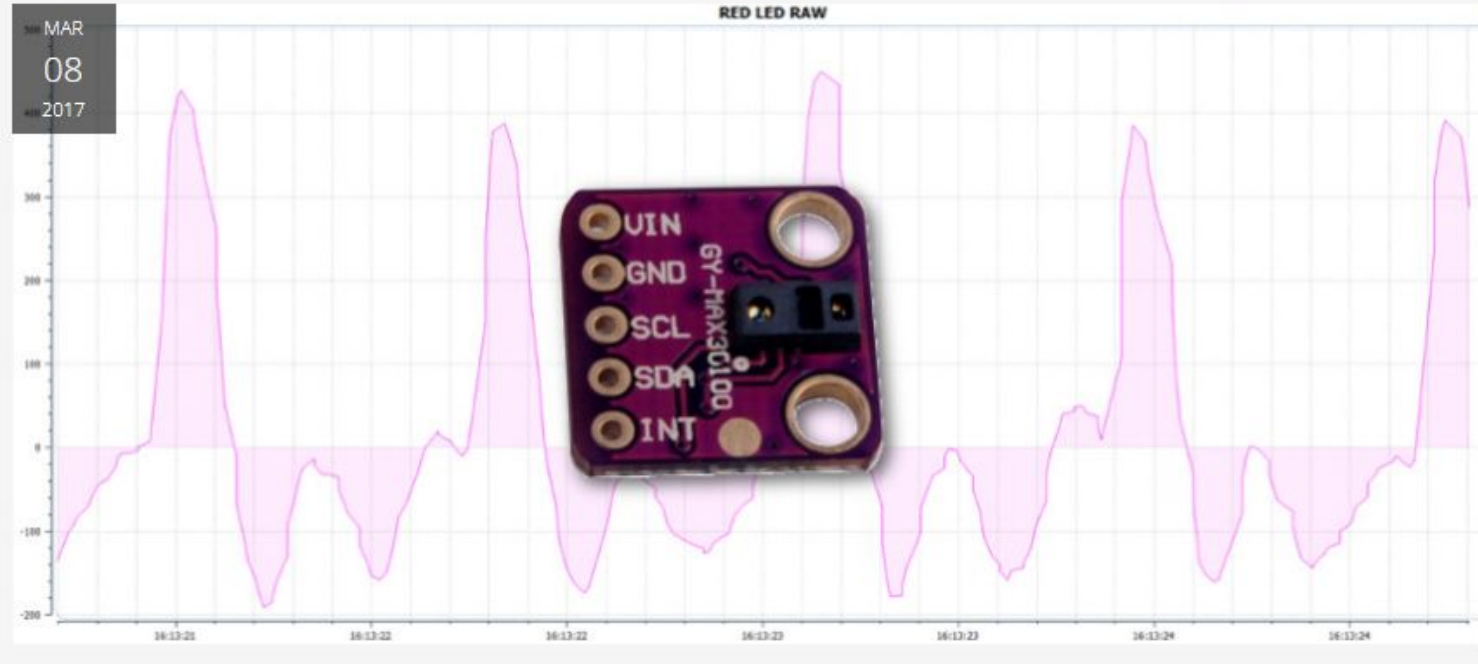


C++ code is EXCEPTIONALLY well documented

Implementing pulse oximeter using MAX30100

by Raivis Stroganovs In: Microcontrollers, Atmel

max30100, pulse oximeter, arduino 120 Comments

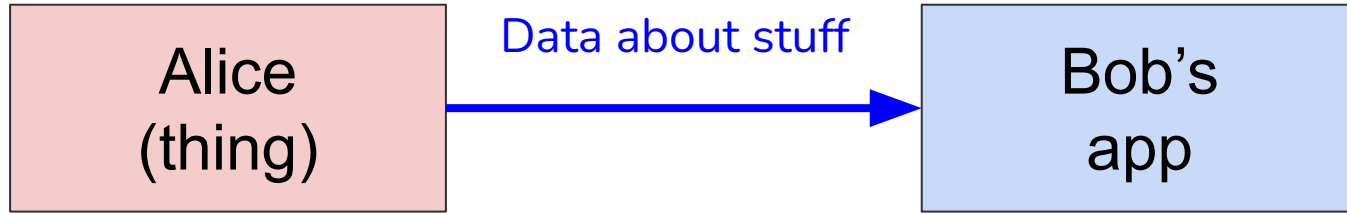


<https://morf.lv/implementing-pulse-oximeter-using-max30100>

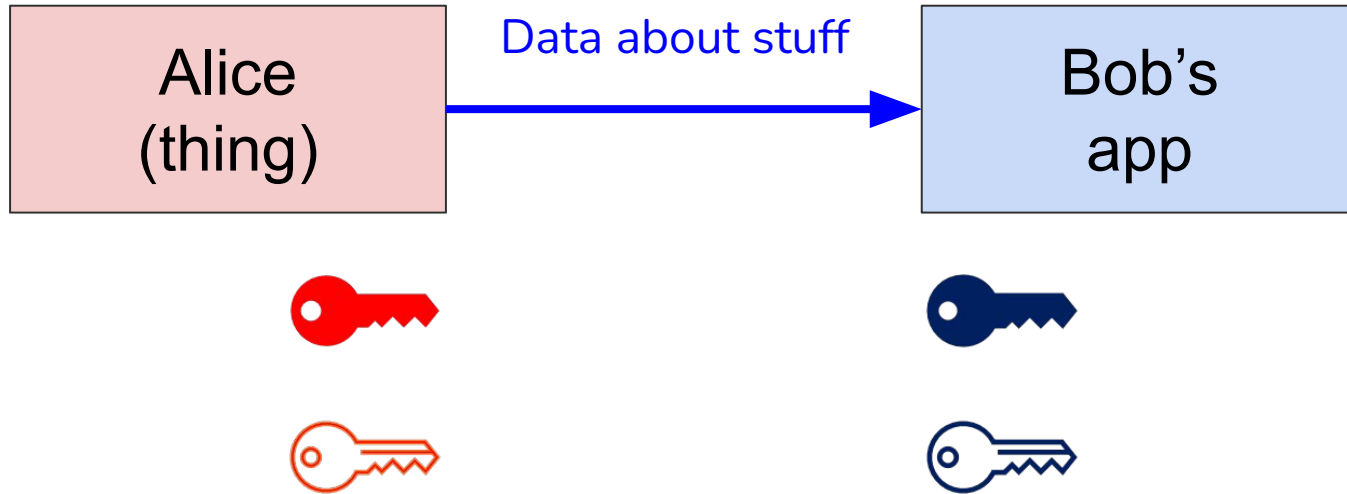


Connectivity, Security, Privacy

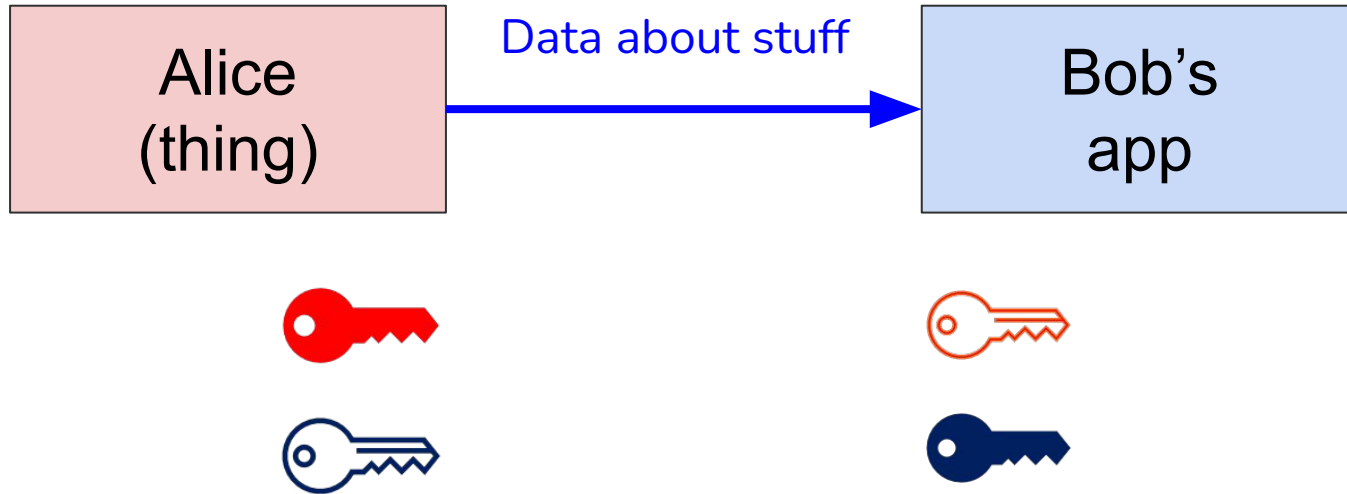
Alice_thing - Bob_person_app



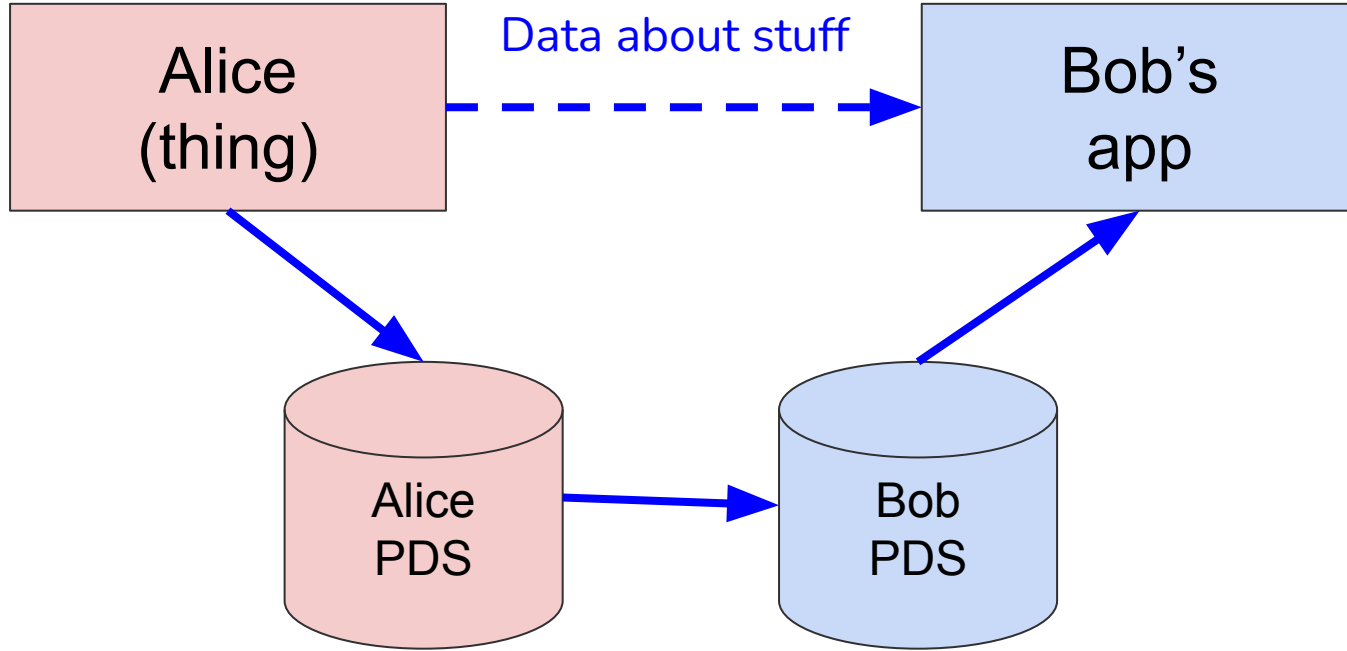
Alice and Bob each have public and private keys



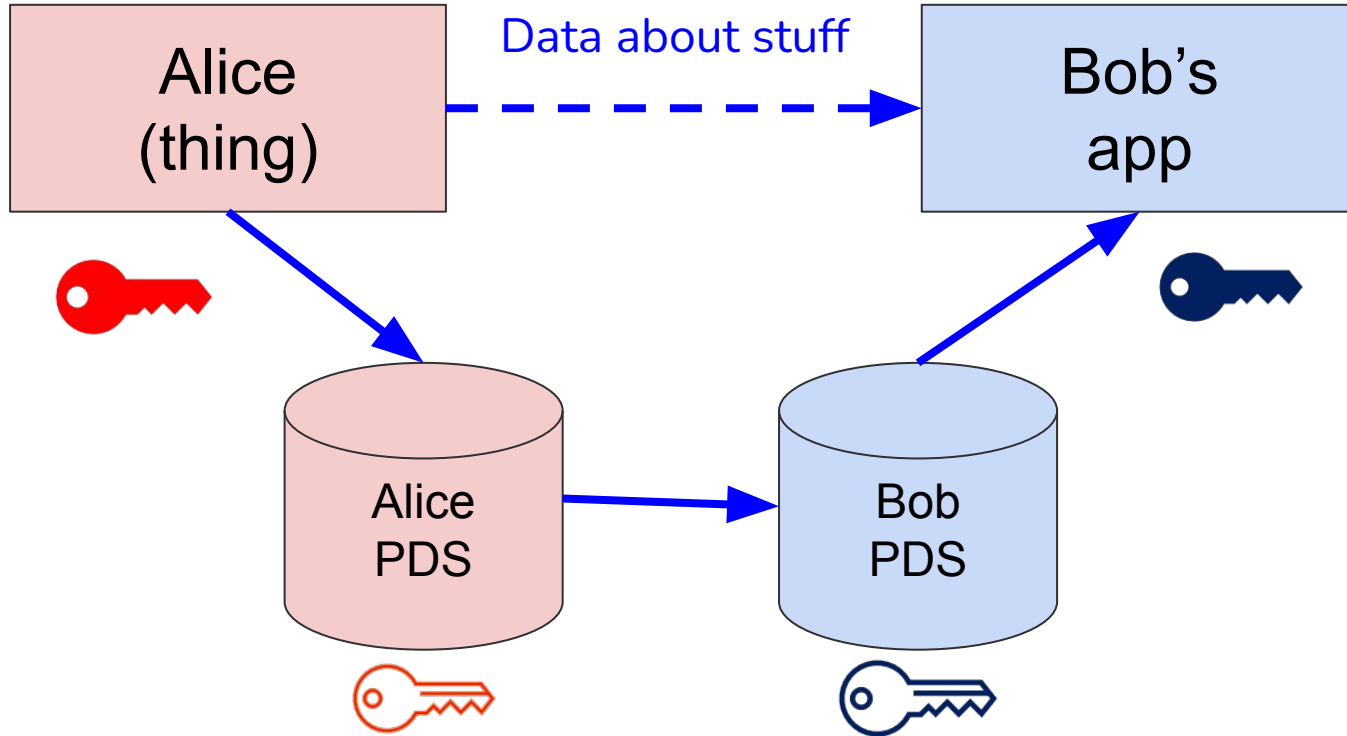
Public keys need to be exchanged



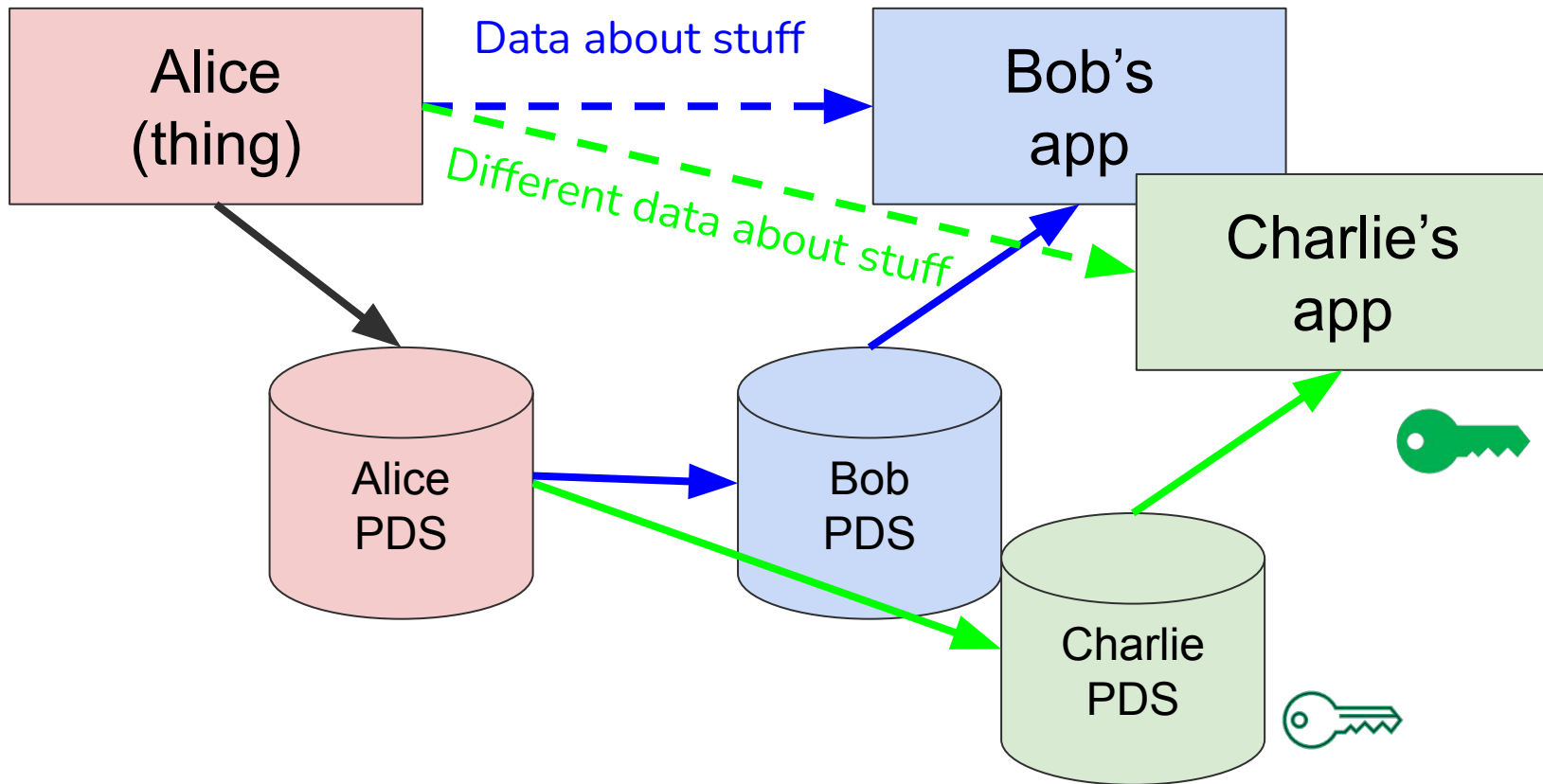
Introduce personal data stores



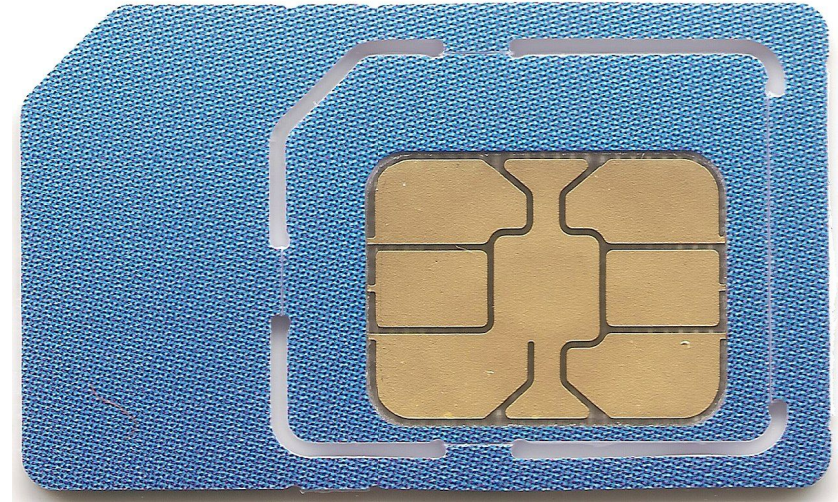
This helps us manage key exchange



Differential Privacy



Where do the keys go?



[Keys by Jay Gooby](#) (CC BY 2.0)



Admin access - Ssh! No ports



Ssh! No ports

ssh no ports provides a way to ssh to a remote linux host/device without that device having any open ports (not even 22) on external interfaces. All network connectivity is out bound and there is no need to know the IP address the device has been given. As long as the device has an IP address, DNS and Internet access, you will be able to connect to it.



<https://github.com/at-sign-foundation/sshnoports>

Review

- Why?
 - ◆ Your Thing, Your Data
- Overview of Mobile World Congress Demo
 - ◆ End to end encrypted data from device (Dart) to app(s) (Flutter) with ability to send different data to different people
- Connectivity, Security, Privacy
 - ◆ Personal data stores simplify connectivity and key management



Thanks for your time

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@cpswan

Get started with the atPlatform



<https://docs.atsign.com/>



<https://github.com/atsign-foundation>



@atsigncompany

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Questions?